

Procedures

1 Introduction

In this document you will find the procedure for some of the exercises seen during the course. The intention of this is to help you into develop math skills and also learn by coping the procedures. I hope this help you.

2 Property of logarithms

1 Reduce the following

3 Solving equations

In this section are some solutions of different types of equations.

Problem 1: Example of exponential equation

We are going to solve the following equation

$$8^{x-2}$$

Since is an exponential function we apply a logarithmic function in order to apply logarithmic properties to manipulate the expression.

Answer : $(e^2, 1)$

4 Modelling with exponential and logarithmic functions

4.1 Critical points

To find the critical point we need to remember that for exponential functions we equate the argument to 0 and then for x . On the other hand, for logarithmic functions we need to equate the argument of the function to 1 and solve for x . Finally, we substitute the value of x into the function and find the value of $f(x)$.

Problem 2: Example of exponential functions

Lets find the critical point of

$$f(x) = \exp[e \ln(x) - 2e]$$

Since the argument is $e \ln(x) - 2e$, lets equate to 0 and solve for x .

$$\begin{aligned} e \ln(x) - 2e &= 0 \\ e \ln(x) - 2e + 2e &= 0 + 2e \\ e \ln(x) &= 2e \\ \frac{1}{e} e \ln(x) &= \frac{1}{e} 2e \\ \ln(x) &= 2 \\ e^{\ln(x)} &= e^2 \\ x &= e^2 \end{aligned}$$

Now that we know the value of x , we can substitute this value into the original equation to find the value of $f(e^2)$.

$$\begin{aligned} f(x) &= \exp[e \ln(x) - 2e] \\ f(e^2) &= \exp[e \ln(e^2) - 2e] \\ &= \exp[2e \ln(e) - 2e] \\ &= \exp[2e - 2e] \\ &= \exp[0] = e^0 \\ &= 1. \end{aligned}$$

Hence the critical point is at $(e^2, 1)$.

Critical point at : $(e^2, 1)$

Problem 3: Example of logarithmic functions

Lets find the critical point of

$$g(x) = \ln [\exp [x^2] - 8].$$

Since the argument is $\exp [x^2] - 8 = e^{x^2}$, lets equate to 1 and solve for x .

$$e^{x^2} - 8 = 1$$

$$e^{x^2} - 8 + 8 = 1 + 8$$

$$e^{x^2} = 9$$

$$\ln [e^{x^2}] = \ln [9]$$

$$x^2 \ln [e] = \ln [9]$$

$$x^2 = \ln [9]$$

$$(x^2)^{1/2} = (\ln [9])^{1/2}$$

$$x^{\frac{2}{2}} = (\ln [9])^{1/2}$$

$$x = \sqrt{\ln [9]}.$$

Now that we know the value of x , we can substitute this value into the original equation to find the value of $f(\sqrt{\ln [9]})$.

$$g(x) = \ln [\exp [x^2] - 8]$$

$$\begin{aligned} f(\sqrt{\ln [9]}) &= \ln \left[\exp \left[\left(\sqrt{\ln [9]} \right)^2 \right] - 8 \right] \\ &= \ln \left[\exp \left[\left((\ln [9])^{1/2} \right)^2 \right] - 8 \right] \\ &= \ln \left[\exp \left[\left((\ln [9])^{2/2} \right) \right] - 8 \right] \\ &= \ln [\exp [\ln [9]] - 8] \\ &= \ln [9 - 8] \\ &= \ln [1] \\ &= 0 \end{aligned}$$

Hence the critical point is at $(\sqrt{\ln [9]}, 0)$.

Critical point at : $(\sqrt{\ln [9]}, 0)$